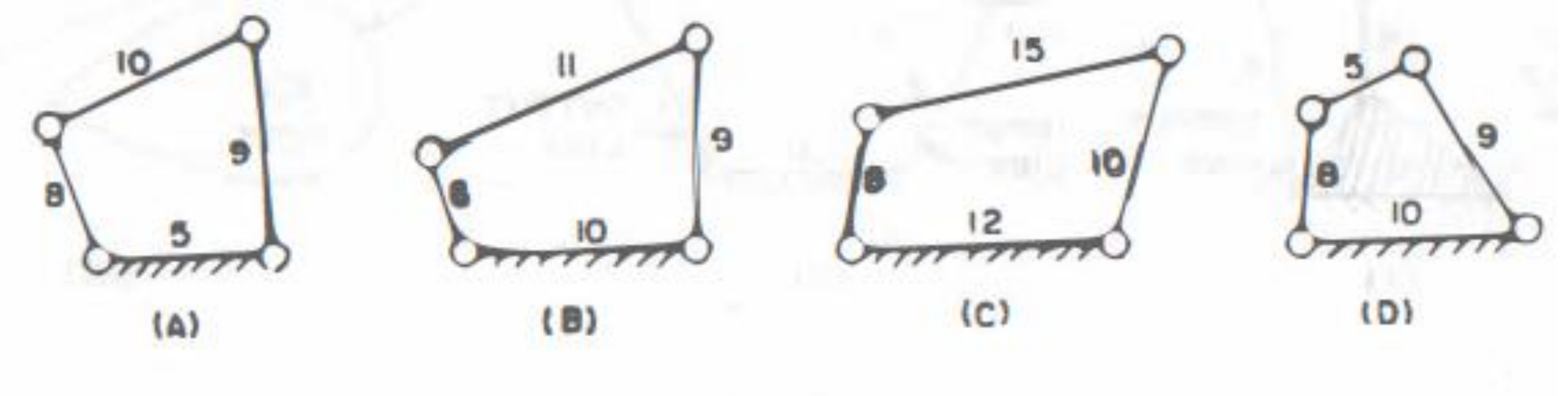
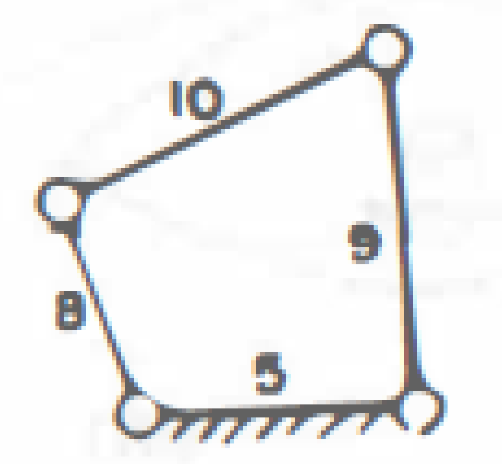
Theory of Mechanism and Machine

(Week-3)

Q 1: *Some four-bar linkages are shown in Fig. where the number indicate the respective link lengths in cm. Identify the nature of each mechanism, i.e. whether (i)double crank, (ii) crank rocker or (iii) double rocker. Give reasons in brief*



- *Data:*
- *A) Length of longest link=l=10, Length of shortest link=s = 5*
- *Length of the other links, p = 8 and q = 9*
- *To identify the nature of each mechanism, Grashoff's law is used. According to this law if* $l+s \leq p+q$

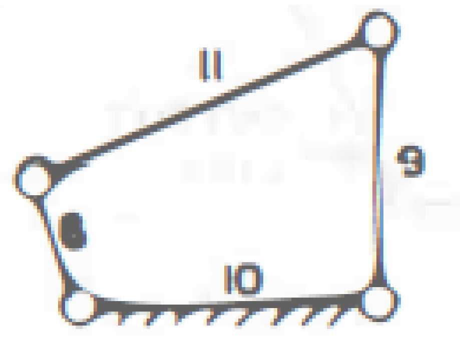


(A)

$l+s = 10+5 = 15$
$p+q = 8+9 = 17$
$l+s < p+q$

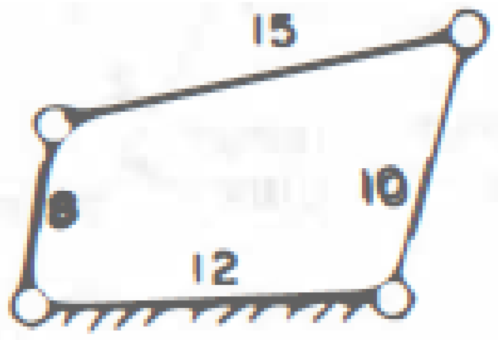
Chain at (A) satisfies Grashoff's law. Since in this case, the shortest link is fixed, the mechanism belongs to the category of double crank mechanism.

- *For chain shown in Fig. (B), we have*
- *l+ s = 11+6=17 and p+q=10 +9 =19*
- *l+s < p+q*
- *Hence chain at (B) satisfies Grashoff's law. Since in this case neither shortest link nor the link opposite to shortest is fixed, the mechanism is of the category of ' 'Crank-Rocker' ' type.*



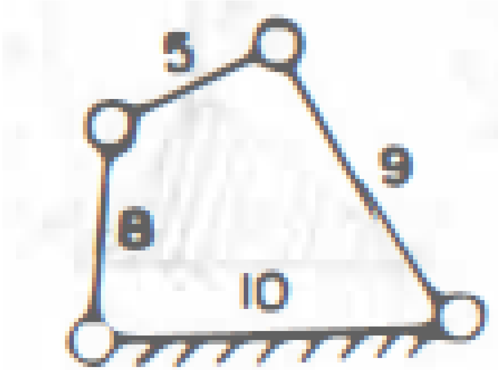
(B)

- For chain shown in Fig. 1C), we have
- $L + s = 15 + 8 = 23$, $p + q = 12 + 10 = 22$
- $l + s > p + q$
- Chain at (C) does not satisfy Grashoffs law. Hence mechanism at (C) is Rocker-Rocker type



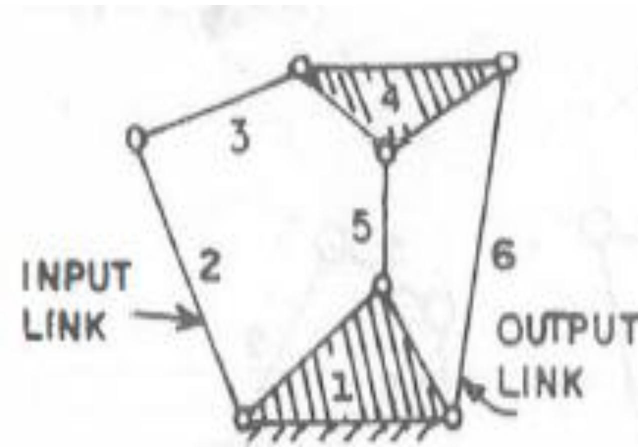
(C)

- *For chain shown in Fig. (D), we have*
- $l + s = 10 + 5 = 15$
- $p + q = 8 + 9 = 17$
- $l + s < p + q$
- *Hence the chain at (D) satisfies Grashoff's law. In this case, neither the shortest link nor the link opposite to shortest link is fixed, hence mechanism belongs to the category of crank-rocker mechanism.*



(D)

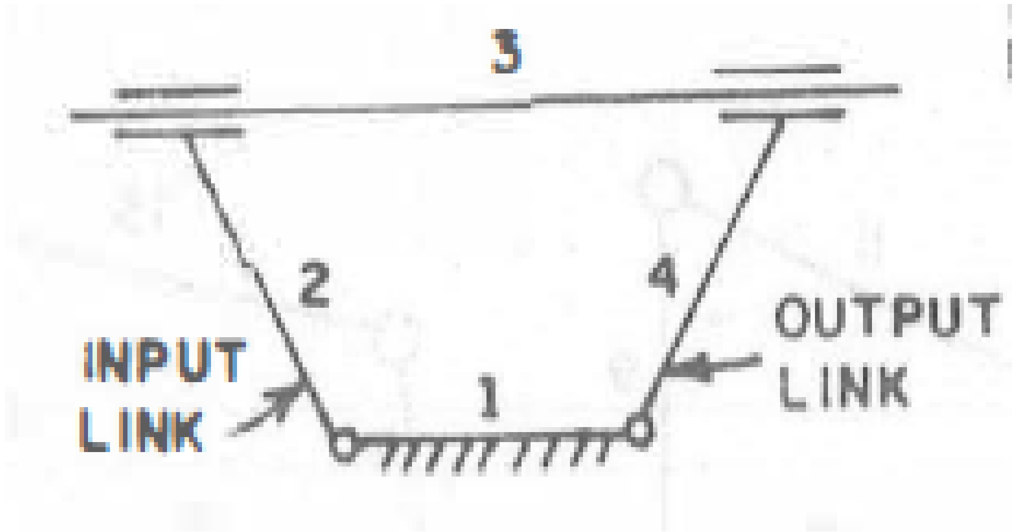
- Q 2: *Examine the mechanism shown in Fig. and indicate the cases where unique relation between the motions of the input and output link exist.1·. Give reasons for your answers.*
- *Case-i*



The Grubler's criterion for planer mechanism is given by
Where F=3(n-1)-2j-h
F= degree of freedom
n=no. of links
j=no. of simple hinges or of degree of freedom= 1
h = no. of higher pair

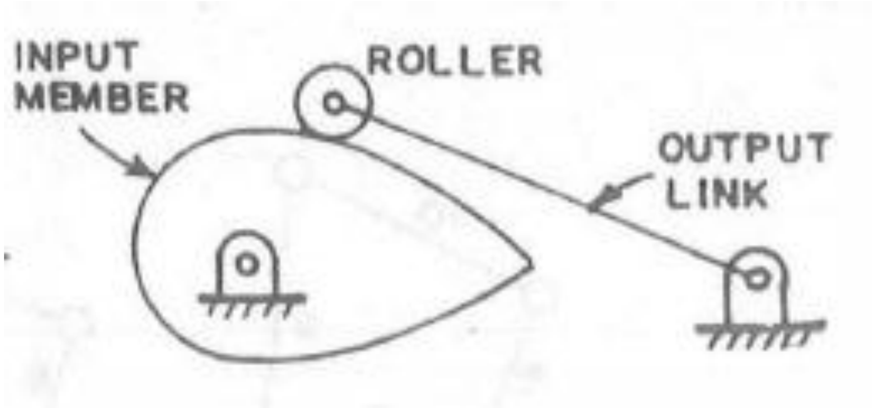
n=6,j=7,h=0
F=3(6-1)-2x7=15-14=1
As in this case, degree of freedom is one, hence unique solution is possible

Case-ii



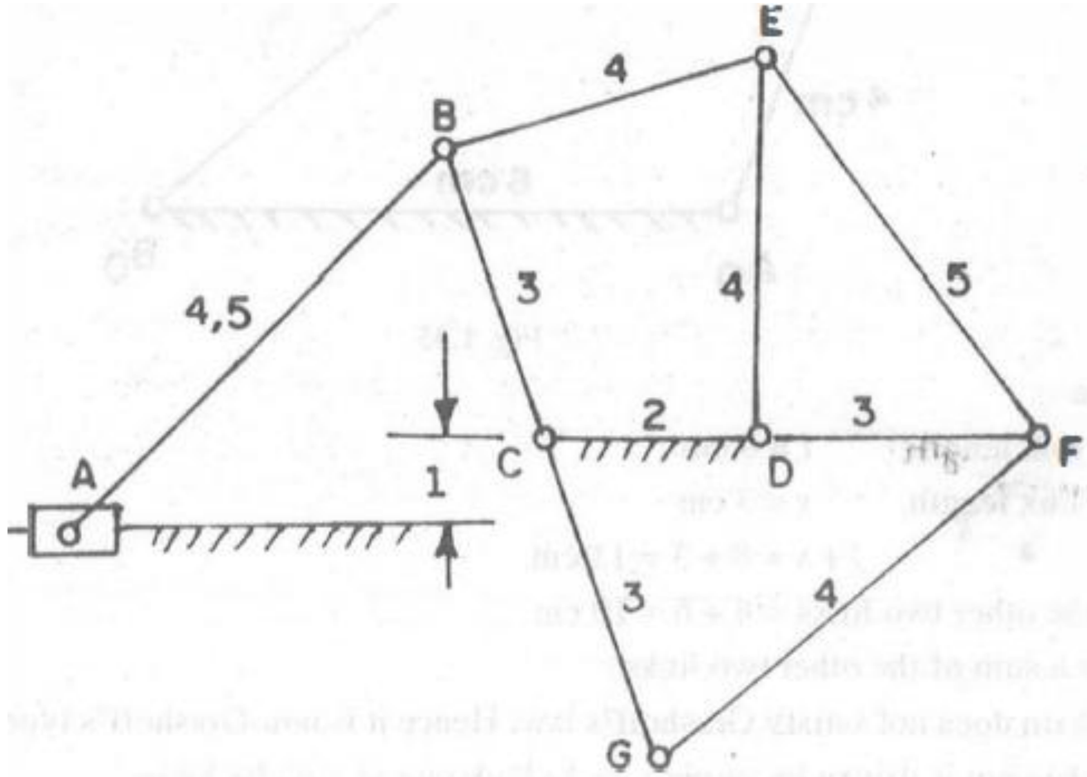
In the mechanism shown in Fig. (ii), there are four links (i.e. links 1 , 2, 3 and 4) and four lower pairs.
But-link 3 is capable of sliding without the help of remaining links.
Hence the mechanism in Fig. (ii) is a locked system .
Also link 3 has redundant degree of freedom

Case-iii



- *The mechanism shown in Fig. (iii), has a roller (carried at the end of output link) which can be rotated without causing any motion in the rest of the mechanism. Thus roller is a link with redundant degree of freedom.*
- *The roller can thus be considered welded to the output link. Hence in this case there arc three links 1,2,3 together with two twining pairs and one higher pair. Thus*
- *n=3, j=2 and h=l ,* $F=3(3-1)-2(2)-I=6-4-1=1$

Q 3: The Fig. shows a plane mechanism with link lengths given in some unit. If slider A is the driver, will link CG revolve or oscillate Justify your answer.



Solution:

The mechanism involves three sub-chains

(i) A four-bar chain CBED,

(ii) A four-bar chain CDFG and

(iii) A slider- crank chain ABC.

i) In four-bar chain CBED, longest link length, $l=4$ units and shortest link length $s=2$ units.

The remaining two links are having lengths as 3 and 4 units.

$$l + s = 4 + 2 = 6$$

Sum of lengths of other two links = 3 + 4 = 7

$$l+s < \text{sum of lengths of other two links}$$

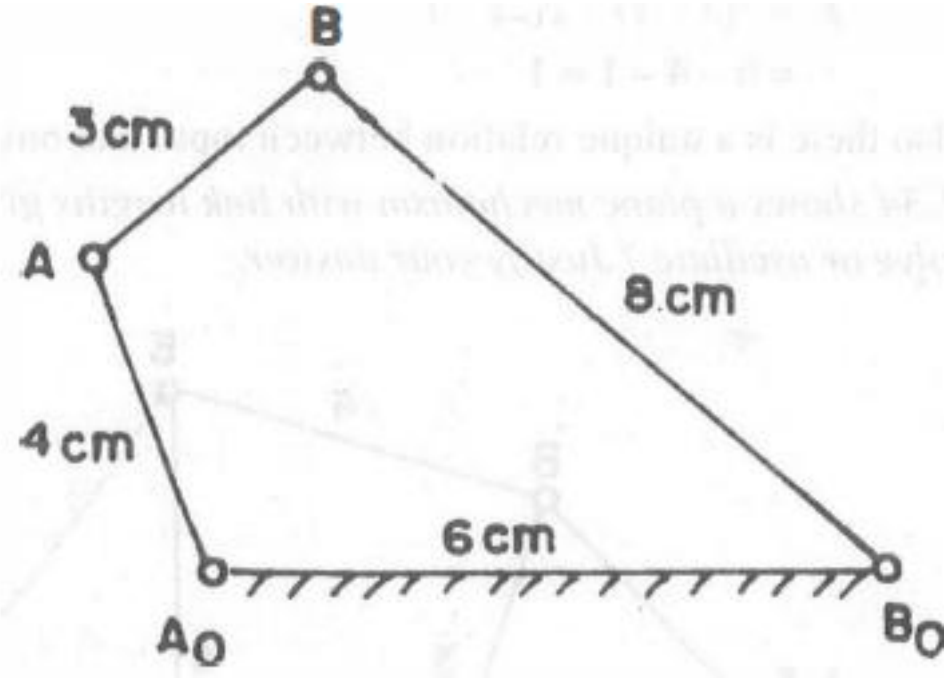
Hence the above chain satisfies Grashoff's law and hence it is a Grashoff's chain. In this case as the shortest link CD is fixed, it represents a double crank mechanism *(ii) In four-bar chain CDFG, longest link length, L=4 and shortest link length, s=2 units. The remaining* two links are having lengths as 3 and 3 units.

l + s = sum of other two link lengths.

Hence this chain also satisfies Grashoffs law and hence it is a Grashoff's chain. Here also the shortest link is fixed and hence it represents a double crank mechanism.

(iii) The third sub-chain is the slider-crank chain ABC.

Since length AB = 4.5 units is greater than the length *BC+offset -1 unit, the crank CB can revolve fully.* Since *CDEB and CDFG are double-crank mechanism,* *Q 4: The mechanism as shown in Fig. is driven by turning AoA . Find out geometrically the maximum and minimum transmission angles.*



Longest link length, l=8cm, Shortest link length, s=3cm

L + s = 8 + 3 = 11 cm

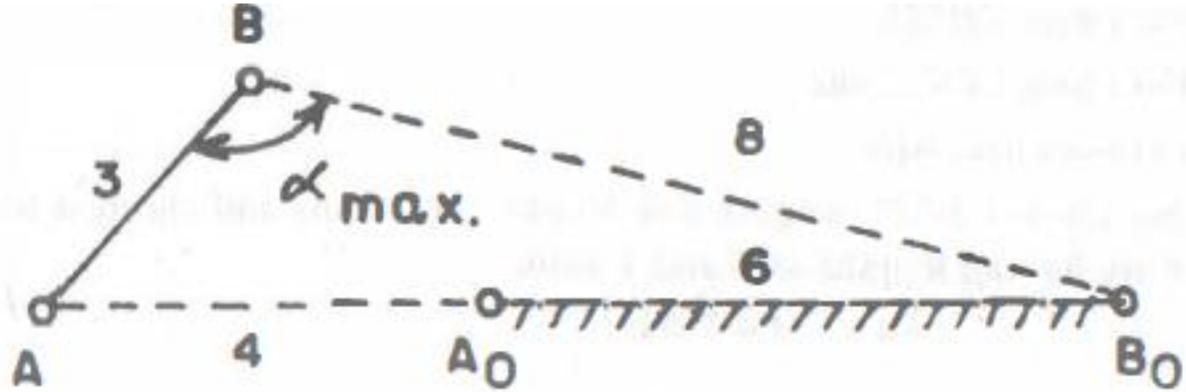
Sum of the other two links = 4+6=10 cm

l + s > sum of the other two links

Hence chain does not satisfy Grashoff's law. Hence it is non-Grashoff's type.

The mechanism is driven by turning A_0A clockwise or anticlockwise

i) When the link AoA is turned anti-clockwise, the link AAo will lie along B_0A_0, This will give the maximum transmission angle (i.e. α_{max}) as shown in Fig.



In triangle B_0AB, the value of α_{max} is given by,

$$AB_0^2 = AB^2 + BB_0^2 - 2 \times AB \times BB_0 \times \cos \alpha_{max}$$

$$(6+4)^2 = 3^2 + 8^2 - 2 \times 3 \times 8 \times \cos \alpha_{max}$$

$$100 = 9 + 64 - 48 \cos \alpha_{max}$$

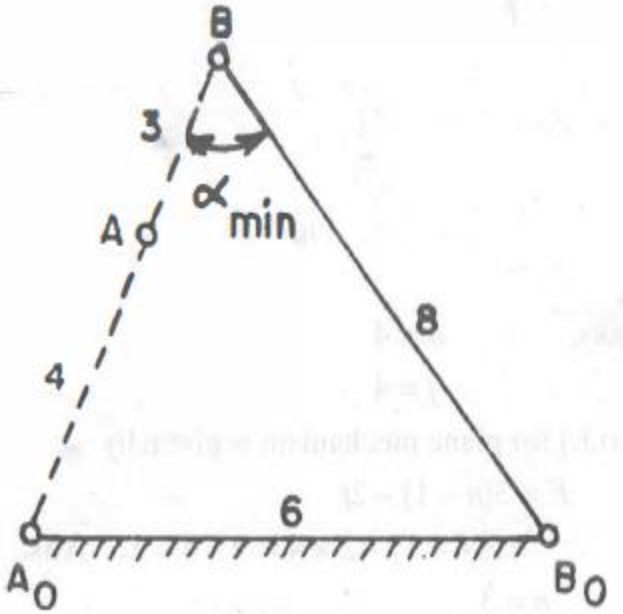
$$= 73 - 48 \cos \alpha_{max}$$

$$\cos \alpha_{max} = \frac{73 - 100}{48} = \frac{-27}{48}$$

$$\therefore \quad \alpha_{max} = \mathbf{124.23^\circ}. \quad \textbf{Ans.}$$

(ii) When link A_0A in Fig. is turned clockwise, the extreme possible position of link A_0A will be when A_0A and AB become collinear as shown in Fig.

Now from triangle B_0A_0B, the minimum transmission angle (α_{min})as



$$A_0B_0^2 = A_0B^2 + B_0B^2 - 2 \times A_0B \times B_0B \times \cos\alpha_{min}$$

$$6^2 = (4+3)^2 + 8^2 - 2 \times (4+3) \times 8 \times \cos\alpha_{min}$$

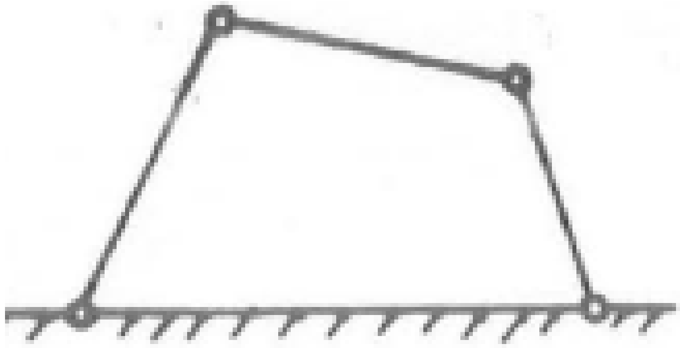
$$36 = 49 + 64 - 112\cos\alpha_{min}$$

$$36 = 113 - 112\cos\alpha_{min}$$

$$\cos\alpha_{min} = \frac{113-36}{112} = \frac{77}{112} = \frac{11}{16}$$

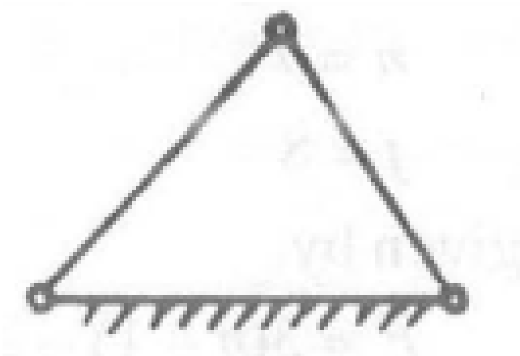
$$\alpha_{min} = 46.58°. \quad \textbf{Ans.}$$

Practice Problem:



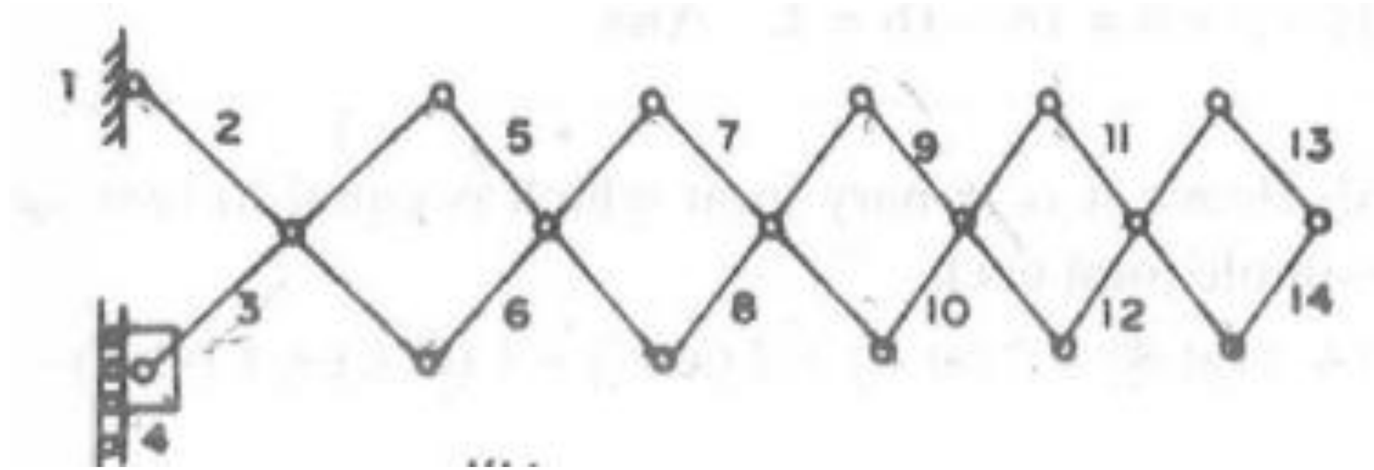
(t) Here, no. of links, n = 4
No. of simple hinges, j = 4
Degree of freedom (d.o.f.) for plane mechanism is given by

$$F = 3(n - 1) - 2j$$
$$= 3(4-1)-2 \times 4=9-8 = 1 .$$



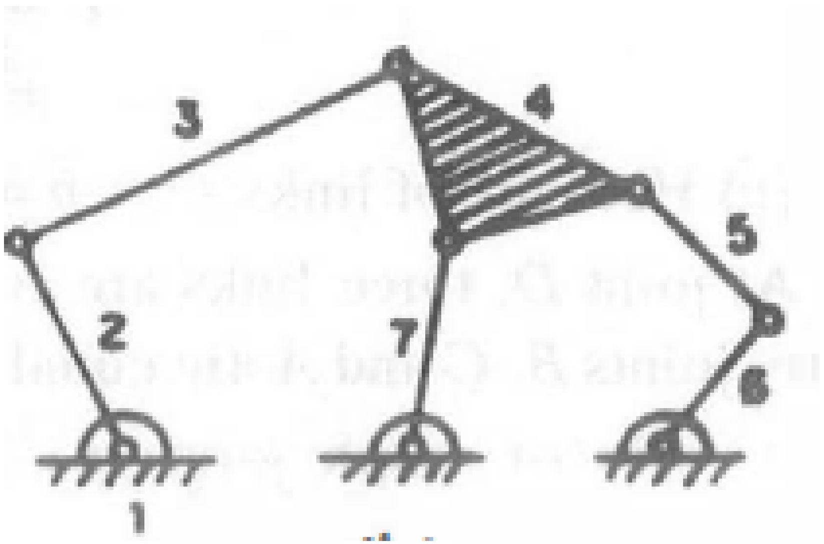
Here, no. of links, n=3
No. of simple binge, j=3
Degree of freedom is given by,

$$F = 3.(n-1)-2j$$
$$= 3(3-1)-2 \times 3=6-6=0.$$



Here, no. of links, n=14
No. of simple hinge, j=18
No. of higher pair, h=1
Degree of freedom is given hy,

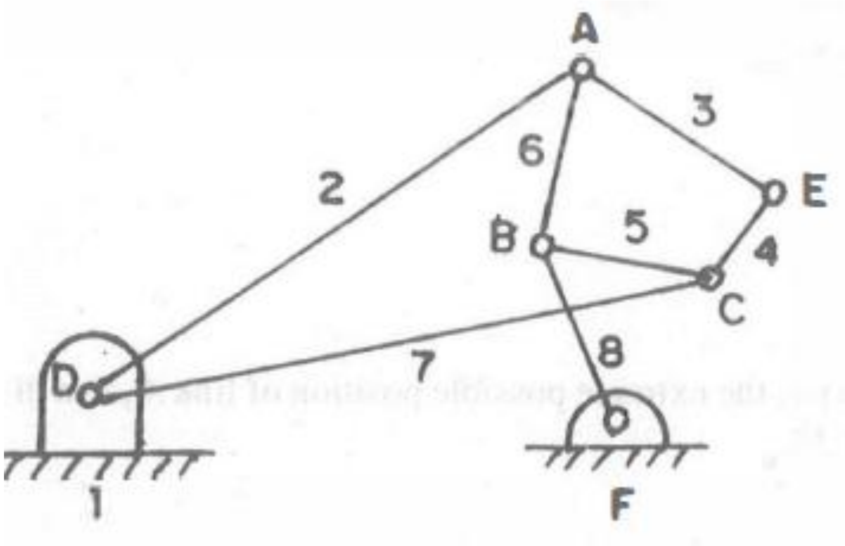
$$F = 3(n - 1) - 2(j) - h$$
$$= 3(14 - 1) - 2(18) - 1 = 39 - 36 - 1 = 2.$$



No. of links, n = 7
No. of simple hinge, j = 8
Degree of freedom is given by,

$$F = 3(n - 1)-2j$$

$$=3(7-1)-2x8 =18-16=2.$$



Here no. of links, n = 8

At joint D, three links are connected. Hence it is ternary joint which is equal to two simple joint

Similarly joints B, C and A are equal to two simple joint each.

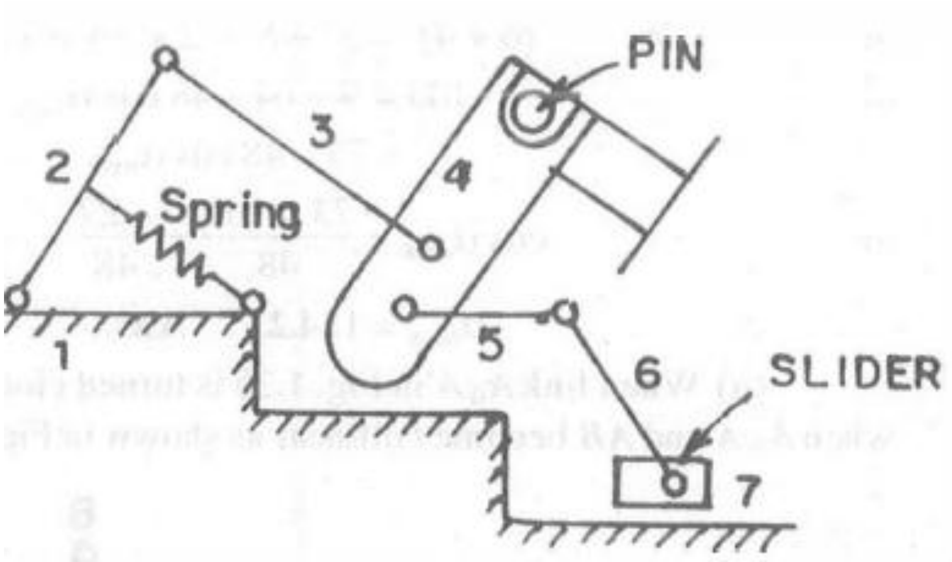
No. of simple joints,

j=2 (at D)+2 (at B)+2 (at A) 1+2 (at C)+I (at E)+I (at F)=10

Degree of freedom is given by,

$$F = 3(n - 1) - 2j$$

$$= 3(8 - 1) - 2x10 = 21 - 20 = 1 .$$



No. of links, n=7
No. of simple joints ,j=7
No. of higher pair, h=1
Degree of freedom is given by,

$$F = 3(n-1)-2(j)–h$$

$$=3(7-1)-2 \, x7-1=18 -14-1=3.$$

Home Work:

- *Extend Gribler's criterion for planer mechanism to obtain the degree of freedom of a space mechanism as:*
- $F = 6(n - 1) - 5g - 4C - 3s$
- *where g = total number of sliding pair,*
- *C=total number of cylindrical pair,*
- *s=total number of spherical pairs and*
- *n=total number of links in the mechanism*